

Fortes Huang

Staff Frontend Architect · Realtime & AI-Assisted Delivery

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SUMMARY

Staff-level Frontend Architect for high-concurrency realtime platforms. I treat frontend architecture as a system-cost and delivery-risk problem: how tabs, sockets, and refetches behave under load decides how expensive a product is to run. I own realtime dataflow coordination and multi-tenant white-label delivery on Vue 3 / TypeScript, and I design AI-assisted delivery workflows that keep correctness and ownership with the team. I also bring product, design, and QA into those workflows, so AI adoption raises real delivery throughput rather than code volume.

SIGNATURE WORK

- **Realtime cost-governance model** — separated a high-frequency data plane (market-tick streams) from a low-frequency control plane (leader election, snapshots, jittered refetch), so N open tabs and windows elect a single realtime owner instead of each opening its own socket and triggering refetch storms. This reframes per-client realtime load from an SRE / cloud-bill problem into a frontend coordination problem.
- **AI delivery-governance framework, taught across functions** — built on a simple premise: AI raises Production Capacity but not Review or Validation Capacity. I codified it into contract / implementation / delivery gates, agent guardrails and skills, and traceable handoffs, then packaged it as a course and reusable kit and rolled it out to product, design, QA, and engineering.
- **Widely-referenced technical writing** — my article on AI adoption in engineering ranks among Google's top results for 「AI 導入工程」 (AI adoption in engineering teams); others cover frontend architecture, AbortController, and TanStack Query.

CORE COMPETENCIES

- **Frontend Platform & Architecture** — multi-tenant / white-label systems, design-token theming, build-time skin isolation, dynamic routing, RBAC permission engines.
- **Realtime Dataflow & Cost Governance** — data-plane / control-plane separation, BroadcastChannel leader election, cross-tab coordination, refetch-storm control, socket lifecycle.
- **AI-Assisted Delivery Governance** — contract-first workflows, review / delivery gates, validation-capacity design, agent guardrails & skills, cross-functional enablement.
- **Performance & Correctness** — heap profiling & budgets, async update strategy, monetary precision, timezone & reporting correctness.
- **Engineering Leadership** — engineering standards, mentorship, cross-team work with Backend / SRE / Design / PM / QA.
- **Stack** — Vue 3, TypeScript, Vite, Pinia, TanStack Query, Zod + ts-rest; Rust (supporting services & tooling); Markdown + Mermaid docs-as-code.

EXPERIENCE

Staff / Lead Frontend Engineer · Frontend Architect — XIHUI Co., Ltd.

Sep 2024 – Apr 2026

White-label cryptocurrency exchange · million-level traffic, high-concurrency workloads

- Frontend architecture owner — system design, technical decisions, and long-term evolution of the platform.
- Designed core frontend infrastructure: Permission Engine (RBAC), dynamic routing, QuerySync, Time Engine, and Precision Engine.
- Built the realtime coordination layer (data-plane / control-plane split, BroadcastChannel leader election, jittered refetch), removing duplicate sockets and refetch storms across multi-tab / multi-window sessions and containing realtime cost at scale.

- Architected a multi-tenant delivery system (dynamic bundling + design-token theming) and a Figma Design-to-Code MCP pipeline that auto-generates design tokens, shortening the design-to-implementation gap.
- Established architecture blueprints, engineering standards, and contract-first API workflows (Zod + ts-rest) to cut integration errors and long-term coupling.
- Reduced JavaScript heap usage by ~40% through a permission-data-model redesign, validated with repeated heap-snapshot analysis.
- Led AI-assisted delivery governance: tuned agent rules and skills (contract-review, phase-verify, pr-review, security-boundary-review, handoff-maintainer) to work inside bounded contexts, with guardrails and contract / implementation / delivery gates requiring every AI output to declare its correctness definition, owner, and shipping evidence.
- Drove cross-functional adoption across PM, design, and QA: packaged the workflow as a course and reusable kit (paired Claude / Codex examples, Linear handoff workflow), framed around Production / Review / Validation Capacity so validation kept pace with AI's faster output.

Senior Frontend Engineer — Tianxin Technology Co., Ltd.

Nov 2023 – Jun 2024

Consumer lifestyle platform (Wota Lifestyle)

- Delivered core user-facing features and stabilized key workflows within an established codebase, extending functionality while keeping consistency with existing architectural patterns.
- Resolved cross-flow integration issues and edge cases with design and backend, improving usability and delivery reliability ahead of launch.

Senior Frontend Engineer — Ascella Technology Co., Ltd.

Mar 2023 – Sep 2023

Design System implementation

- Led the practical implementation of a Design System, turning design principles into reusable components and tokens and making it enforceable in the codebase rather than conceptual.
- Partnered with design leadership to align design intent with engineering reality and set foundational conventions for component structure, styling, and usage.

Senior Frontend Engineer — Vinnovation Network Co., Ltd.

Jul 2022 – Mar 2023

Large-scale content platform

- Built and maintained the frontend for a content platform with complex user interactions.
- Designed a client-side image caching / delivery mechanism (IndexedDB) plus CDN image subscription and Base64 transformation for offline-friendly, sequential reader-style rendering on low-bandwidth and unstable devices.

Senior Frontend Engineer — Hlwtech Co., Ltd.

Jun 2021 – May 2022

Online gaming platform · multilingual systems

- Built user-facing and multilingual management systems for an online gaming platform.
- Drove adoption of Git-based workflows (self-hosted GitLab on CentOS 7, replacing legacy SVN); designed a JSON-based i18n architecture and a CSV ↔ JSON tooling pipeline to cut translation errors and iteration cost.

WRITING, SPEAKING & COMMUNITY

- **Speaker, Vue Taiwan Meetup (2026)** — “Realtime ownership: control plane, leader election, and taming refetch storms.” Open-source reference implementation: [Frontend](#) · [Backend](#) · [Slides](#) · [Live guide](#).
- **Featured article** — [“AI adoption in engineering teams: the real bottleneck is PR review and validation responsibility”](#) — ranks in Google’s top results for 「AI 導入工程」. More at hackmd.io/@FortesHuang.

EDUCATION

Tungnan University — Department of Information Management

2006



Meetup live guide



AI-in-engineering article